

UrbanCart E-Commerce Analysis

Executive Summary | Data spanning Jan 2022–Dec 2024 | 2,500 customers · 300 products · ~9,000 orders

Leadership asked three questions: which customers are worth investing in, which products are actually profitable, and whether the company's own data can be trusted. The analysis answers all three — and surfaces one urgent correction to how revenue is currently reported.

71%

Revenue has been overstated by 71% in reporting that includes cancelled, returned, and pending orders alongside completed ones (\$12.97M reported vs. \$7.57M actually earned). Any decision built on raw order totals — revenue targets, inventory, staffing — was likely built on a number nearly double the truth. **Fix: filter all reporting to completed orders only, immediately.**

Top Findings

- **A small group drives nearly half of revenue.** “Champions” (23% of buyers) generate 48% of revenue (\$3.63M) — and can only be identified by behavior, not demographics.
- **Revenue is highly seasonal.** November (\$273.7K avg.) outsells February (\$151.7K avg.) by ~80%. Flat demand planning misallocates budget.
- **The best category by profit isn't the best by revenue.** Books & Media earns the highest margin (77.95%) despite lower raw sales than other categories.
- **Star ratings don't predict loyalty.** Rating-to-repeat-purchase correlation is ~0.036 — essentially zero.
- **One product is nearly certain to stock out.** “Those Non-Fiction” has 0 units on hand against a 96.5% simulated stockout probability.

Top Recommendations

- Filter all revenue reporting to completed orders only — treat as an immediate data-governance fix.
- Reorder “Those Non-Fiction” now; review 4 other flagged products on a normal cadence.
- Build retention programs triggered by behavior (RFM score), not demographics, around the Champions segment.
- Shift Q1 marketing/staffing budget toward Q4 to match the seasonality pattern.
- Re-evaluate rating-driven loyalty campaigns; test price and cadence levers instead.

Findings are based on the company's cleaned and reconciled order, customer, product, review, and web-session data (2022–2024). Full methodology, data-quality notes, and detailed analysis are available in the accompanying project report.